THE THING ON THE FRONT OF YOUR ROBOT

How the ultrasonic sensor works

It makes a click too high for you to hear, then times how long the echo takes to come back. **Everything else is arithmetic.**



THEN IT DOES ONE SUM

time × 343 ÷ 2

Sound covers 343 meters every second. Halve it, because the click went out *and* came back.

1 Two eyes, two jobs

One is a tiny **speaker**. The other is a tiny **microphone**. That is all it is.

2 It makes a click you cannot hear

A short burst — about **12 vibrations, at 40,000 a second**. Your ears stop near 20,000, so the room sounds silent. It is not.

3 The click bounces back

Exactly like shouting at a cliff and hearing yourself a moment later. Just too fast and too high for you to notice.

4 The robot checks its clock

How long did that take? The only thing it ever measures. It has never measured a distance in its life.

THE THING WORTH REMEMBERING

It never measures a distance. It measures **time**, and then multiplies. Every number it has ever shown you was worked out, not seen.

A SHORTCUT, AND A WAY TO CHECK YOURSELF

343 ÷ 2 is **171.5**, so you can do it in one step: **distance = time × 171.5**. Which gives a rule you can keep in your head — **one thousandth of a second is about 17 cm.**

And if you ever forget the ÷ 2, the robot thinks every wall is **twice as far away as it is**. A wall at 30 cm reports 60.

Turn the wall and it stops working. Suddenly.

Put a board 20 cm away, square on. Read the screen. Now turn the board a few degrees at a time and read it again at each step.

Square on

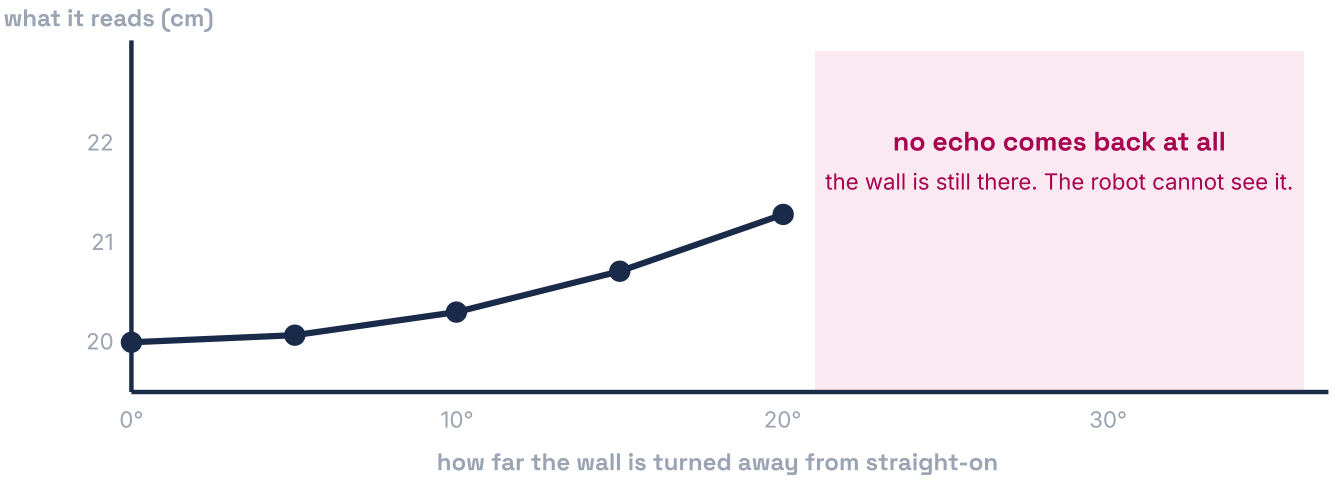


the echo comes straight back

Turned away



it leaves sideways and never returns



Up to about 15°

The number creeps up by a few millimeters. Not enough to notice, and not enough to matter.

Past about 20°

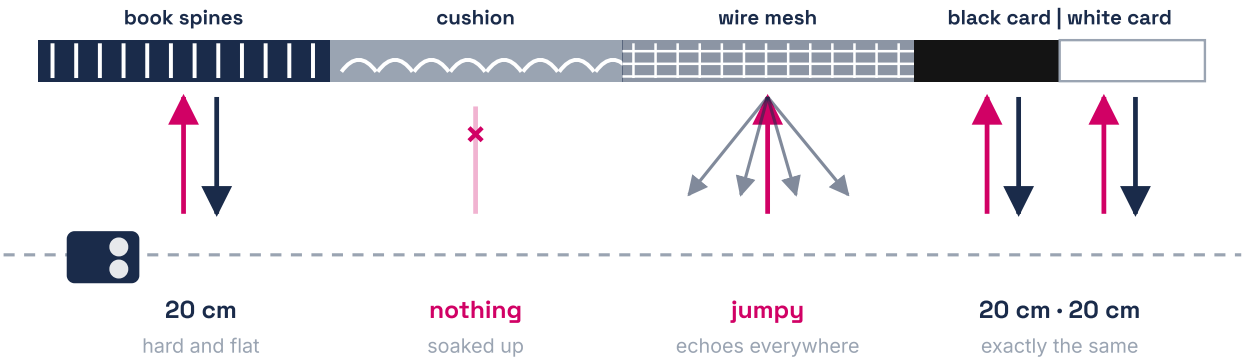
Nothing comes back at all. Not a wrong number — no number. The board is still there.

WHY

The click bounces off a flat surface the way a ball bounces off a wall. Hit it square on and it comes straight back to you. Hit it at an angle and it leaves sideways and never returns. **This is a cliff, not a slope** — which is why a robot following a wall is fine, fine, fine, and then completely lost.

Some things send the click back. Some do not.

Line a stretch of wall with different surfaces. **Predict which ones the robot will lose**, then drive past and mark the floor where it actually fails.



WHY THE CUSHION WINS

Hard flat things bounce almost all of the click straight back. Soft fluffy things let it **sink in and turn into heat** instead. Nothing returns, so the robot decides there is nothing there — and drives into it.

TWO DIFFERENT WAYS TO FAIL

The cushion fails because it **soaks the click up**. An angled wall (page 2) fails for the opposite reason — it is hard and flat and bounces back **almost all** of the click, just sideways, away from the microphone.

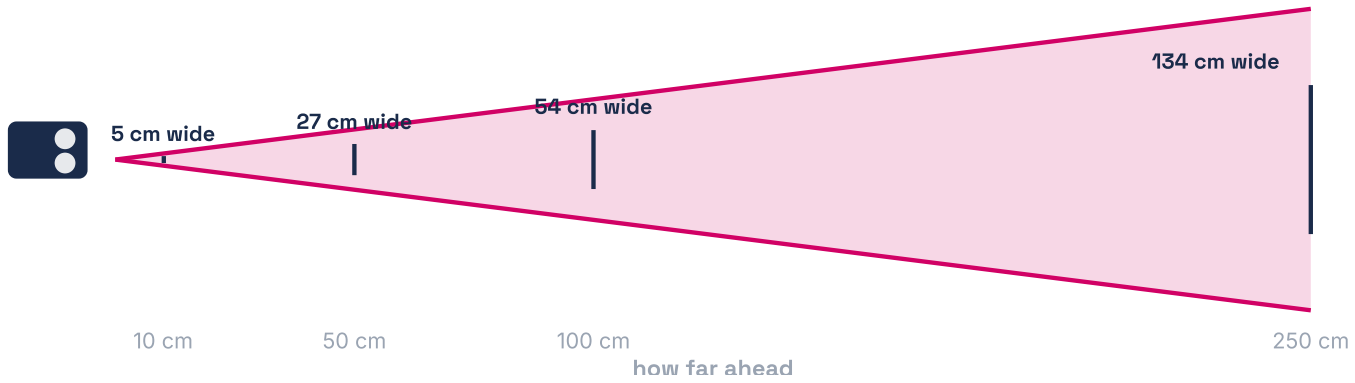
So the rule is not “it must not absorb.” It is **it must send the click back at you.**

THE ONE NOBODY PREDICTS

Black and white read **exactly the same.** Colour is something **light** does. A sound has no way of knowing what colour anything is.

It is not looking at a point. It is looking at a patch.

The click spreads out as it travels, like the beam of a torch. The further away something is, the wider the patch of room the sensor is listening to.



ROUGHLY

The patch is about **half as wide as the distance**. At 50 cm it is listening to about 27 cm of room. At 2 metres it is listening to more than a metre.

WHAT THIS EXPLAINS

A gap in a wall vanishes when you back away from it. Close up, the patch is narrow enough to fit inside the gap and the click sails through. Further back, most of the patch is still hitting solid wall — **and whatever comes back first wins.**

There is a nearest and a furthest. Both have reasons.

The sensor works between about **3 cm and 250 cm**. Outside that it is not being difficult — it physically cannot.



Too close: under 3 cm

The echo gets back in less than **two ten-thousandths of a second** — while the speaker is still buzzing from making the click. The microphone cannot hear a whisper over its own shout.

Too far: past 250 cm

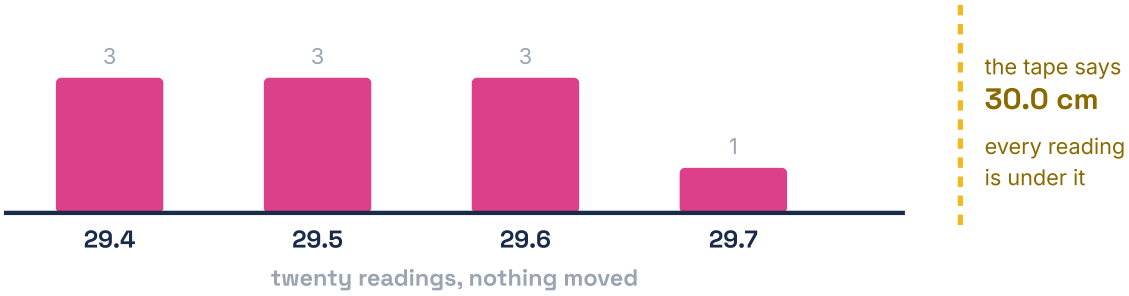
The click spreads out and fades as it goes. By 2.5 metres and back, what returns is too faint for the microphone to pick out at all.

DESIGN RULE

Never mount it where something can get closer than 3 cm. No amount of clever programming fixes a blind spot.

Take twenty readings without moving anything.

Put the robot and a box down and **do not touch either one**. Measure the real gap with a tape. Now read the screen twenty times and write every number down.



twenty readings, nothing moved

They jump around

Up a bit, down a bit, no pattern. **Taking an average sorts this out** — and three readings usually gets you as close as twenty does.

They are all a bit low

The tape says 30.0 and every single reading is under it. **An average will not fix that.** Something is wrong, and you have to find it.

TWO THINGS THAT PUSH EVERY READING THE SAME WAY

Where you measured from. Your tape started at the bumper; the sensor sits further back. · **A cold room.** Sound travels slower when it is cold, so the echo takes longer and the robot turns that into extra distance. At 10 °C a true 30 cm reads about 30.5.

Five ways it fails. Five things you can do.

Knowing *why* the sensor fails is half the job. **An engineer has to decide what to do about it** — and the answer is almost never better code.

PAGE 2 The surface is angled

Correct your **heading**, not just your distance — a climbing reading usually means you are turning, not drifting. **Sweep instead of staring:** one fixed reading is one chance at the echo. And never write **if no reading → drive forward**.

PAGE 3 The surface is soft

You cannot fix this one with sound — soft and sound-invisible are nearly the same property. Either **detect it by the echo it steals** (a known wall going quiet means something is in the way), or **change the energy**: infrared sees a cushion to about 50–70 cm (**page 8**), touch sees anything but only on contact.

PAGE 4 The patch is wider than the thing

Close in. The patch is about half as wide as the distance, so halving the range halves what else can get into the answer. If you cannot close in, **stop asking it to pick one object out of several** — it reports the nearest and says nothing about the rest.

PAGE 5 Something is closer than 3 cm

Mount it so that cannot happen. Set it back from the front of the robot, or point it where nothing will ever arrive that close. The last few centimetres belong to a touch sensor.

PAGE 6 The numbers wobble, or they all lean

Wobble is cured by **averaging**, and three readings usually get you as close as twenty. A lean is not — find the cause. Check **where you measured from** first; the tape starts at the bumper and the sensor sits further back.

THE RULE UNDERNEATH ALL OF THIS

Every sensor relies on **one** property of the thing it is looking at. Sound needs something hard. Light needs something that reflects. Touch needs contact. **Anything without that property is not badly seen — it is not seen at all.**

That is why real robots carry several. Not spares. **Different blind spots.**

Infrared: a torch and a light meter.

Page 7 said that when sound cannot do the job you have to **change the energy**. This is what that looks like.



1 It shines a light you cannot see

Infrared sits **just past red**, where your eyes stop. Point a phone camera at it — most phone cameras *can* see it, and the LED shows up as a faint purple glow. **Same signal a TV remote uses.**

2 Some of it bounces back

Off whatever is in front, in all directions. Only a little of it comes back towards the sensor.

3 A detector measures how much

Not how long it took. **How bright it is.** That is the only thing this sensor ever measures.

4 Brighter means nearer

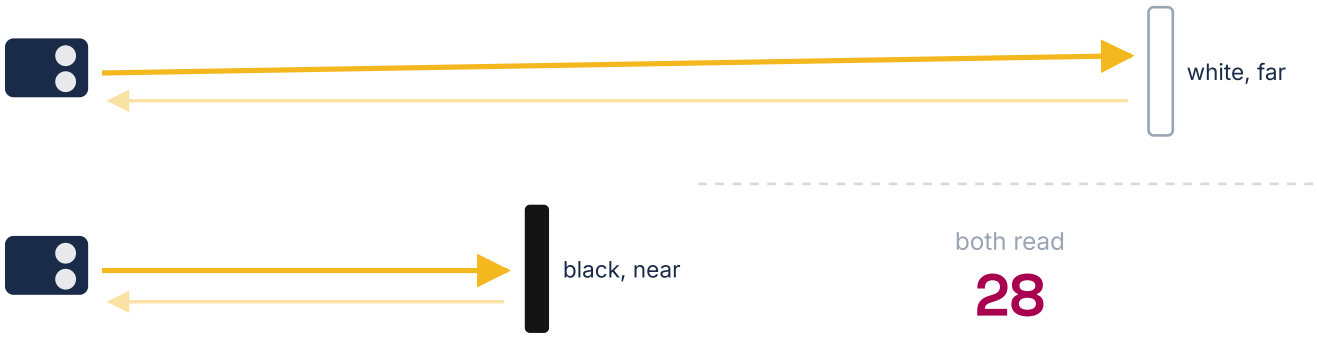
Light **spreads out** as it travels, so the further it goes the less of it finds its way back. It reports **0 to 100** — not centimetres, because it has no way of working them out.

THE WHOLE DIFFERENCE, IN ONE LINE

The ultrasonic **times a journey.** The infrared **judges a brightness.** Everything else about them follows from that.

It cannot tell dark from far away.

A pale wall a long way off, and a black card right in front of it. Both send back the same small amount of light — so the sensor reports the same number for both, and has no way of knowing which it is looking at.



Dark things look distant

Black absorbs the light instead of returning it. A black object close up can read the same as a white one far away.

Shiny things look empty

A smooth surface bounces the light off sideways, like the angled wall on page 2. Nothing comes back, so it reads as nothing there.

TWO MORE THINGS THAT CATCH IT OUT

Sunlight. The sun pours out infrared, and the detector cannot tell the sun's light from its own — so near a window the readings drift. · **Distance.** Light spreads as it travels, so past about **50 to 70 cm** there is too little coming back to measure at all.

Neither is better. They fail at different things.

Put both on the bench, point them at the same four targets, and read them off Port View. **Six of these eight readings you can already predict.**

POINTED AT	ULTRASONIC SAYS	INFRARED SAYS	BECAUSE IT IS…
A cushion	nothing at all	sees it clearly	soft, but pale
A black card at 20 cm	20 cm	a weak number	hard, but dark
A white card at 20 cm	20 cm	a strong number	hard and pale
A wall turned 30°	nothing at all	nothing at all	both bounce away

Ultrasonic

Measures **time** → real centimetres, 3 to 250. Ignores colour completely. Beaten by **soft** and by **angles**.

Infrared

Measures **brightness** → 0 to 100, out to about 50–70 cm. Ignores softness completely. Beaten by **dark**, by **shine** and by **sunlight**.

THE RULE UNDERNEATH ALL OF THIS

Every sensor relies on **one** property of the thing it is looking at. Sound needs something hard. Light needs something that reflects. **Anything without that property is not badly seen — it is not seen at all.**

The cushion that is invisible to one is obvious to the other. Not because one is better, but because **their blind spots do not overlap.**

Everything in here you can test yourself.

Not one of these needs a program. The brick's **Port View** shows the sensor live. The page number is where this document makes the claim — **go and disagree with it.**

With the ultrasonic

EXPERIMENT	WHAT TO DO	WHAT YOU WILL FIND
Find the angle cliff P. 2	Board at 20 cm, square on. Turn it 5° at a time, reading at each step.	Fine to about 15°, then nothing at all. A cliff, not a slope.
Four surfaces P. 3	Book spines, a cushion, wire mesh, black card beside white. All at 20 cm.	The cushion vanishes. Black and white read identically.
Walk the pencil out P. 4	Pencil to the side of the centre line. Walk it away until the robot notices.	Mark each point, join them: you have drawn the cone.
Two objects, one answer P. 4	Books at 20 and 40 cm, both in front. Read. Take the near one away. Read.	It only ever reports the nearer. The far one did not exist.
Find the floor P. 5	Bring a card slowly in from 20 cm towards the sensor face.	It tracks beautifully, then gives up at about 3 cm.
Twenty readings, nothing moving P. 6	Robot and target both still. Read the screen twenty times. Write all of them down.	They scatter. They may also all lean one way.

With the infrared

EXPERIMENT	WHAT TO DO	WHAT YOU WILL FIND
Make the invisible light visible P. 8	Point a phone camera at the sensor while it is running.	The LED glows faint purple on the screen. Same trick on a TV remote.
Dark or distant? P. 9	White card far away; black card near. Move them until both read the same.	Two completely different situations, one number. It cannot tell them apart.
Go to the window P. 9	Fixed distance from a card in the middle of the room. Then by a sunny window.	The number drifts. Nothing moved except the light in the room.
Find its limit P. 9	Walk a card away until the number stops meaning anything.	Somewhere around 50 to 70 cm it stops being useful.

Both sensors, and then the big one.

The three below need both sensors on the bench at once. The last one needs twenty minutes and is worth every one of them.

With both, side by side

EXPERIMENT	WHAT TO DO	WHAT YOU WILL FIND
The cushion, both ways P. 10	Both sensors, same cushion, same 20 cm. Read both off Port View.	One sees nothing. The other sees it clearly.
Black and white, both ways P. 10	Both sensors, black card then white, same distance, nothing else changed.	Ultrasonic: identical. Infrared: completely different.
The angled wall, both ways P. 10	Turn a hard board 30° and read both.	Both fail. Infrared does not fix everything.

The one that puts it all together

EXPERIMENT	WHAT TO DO	WHAT YOU WILL FIND
Radar	Robot on paper marked every 15°. Turn by hand to each mark, read, write it down. Then plot each distance along its own line and join the dots.	A map of the room — and every failure above shows up in it at once.

WHY RADAR IS THE RIGHT LAST EXPERIMENT

A flat wall draws clean. An angled corner drops out. A doorway appears as a gap. Anything past 250 cm simply is not there. **Four failures you chased one at a time, all visible in a single drawing** — and no programming anywhere in it.